Comparative Analysis: Neural Networks, Random Forests, and Naive Bayes for Lyrics-Based Music Genre Classification

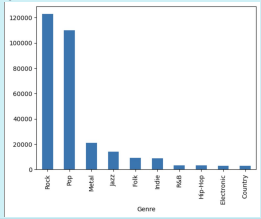
Lily Meyer, Meghna Nimmala, Nidhi Satti

Abstract

- Compared ANN, Random Forest, and Gaussian Naive Bayes to classify 10 music genres from lyrics only.
- Used TF-IDF vectorization for semantic weighting.
- Random Forest performed best (76.17%), followed by ANN (73.68%) and NB (65.65%).
- Results reveal limits of lyrics-only classification; audio features likely needed for higher accuracy.

Introduction

Problem: Most genre classifiers use audio; this project asks: *Can lyrics alone predict genre?*
Goal: Identify which ML model best captures textual/semantic differences across genres.



Methodology

- Data Collection and Preprocessing
- Feature Engineering - TF-IDF
- Model Training + Evaluation

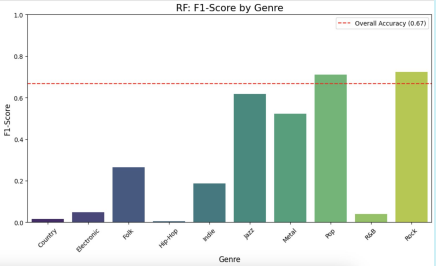
Models

1. Naive Bayes
2. Neural Network
3. Random Forest Classification

Evaluated each model using four metrics:

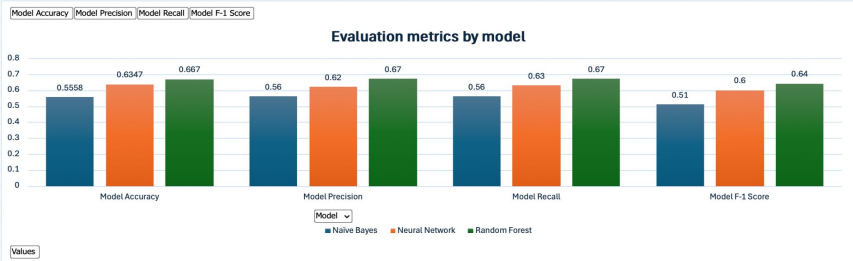
- **Accuracy**: overall prediction correctness
- **Precision**: correctness of predicted genres
- **Recall**: model's ability to detect each genre
- **F1 Score**: balance between precision and recall

Analysis



Results

Overall we can see that Random Forest had the highest accuracy of 67%.
Naive Bayes had the lowest accuracy of 55%.



Conclusion

Lyrics contain clear linguistic patterns that can reliably distinguish between music genres. Random Forest performed the best at capturing these patterns, highlighting that text-based features along can effectively classify songs. This demonstrates the strong potential of NLP methods in music analysis and genre prediction.

Sources

1. Kaggle - Multilingual Lyrics for Genre Classification
2. Scikit-learn Documentation
3. TensorFlow Documentation
4. Jurafsky & Martin, *Speech and Language Processing*